



Hierarchical Linear Models: An Introduction

Marshall Jean
University of Chicago

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Materials and methods contributed by Steve
Raudenbush

Introduction

A. Two-level cross-sectional data, examples:

- Schools assigned at random to treatments with students nested within schools
- Studying social or ethnic inequality: contributions of student background and school segregation

B. Two-level models for repeated measures, examples:

- Children's vocabulary growth during the second year of life
- Items nested within students, an IRT model

C. Three-level models, examples:

- Students nested within classrooms within schools
- Repeated measures within students within schools

Introduction (cont.)

D. Cross-classified models

- Students nested within neighborhoods and schools
- Repeated measures on students crossed by teachers

E. Four-Level Models

- Students within classrooms within schools within countries
- Students crossed by teachers who are nested within schools

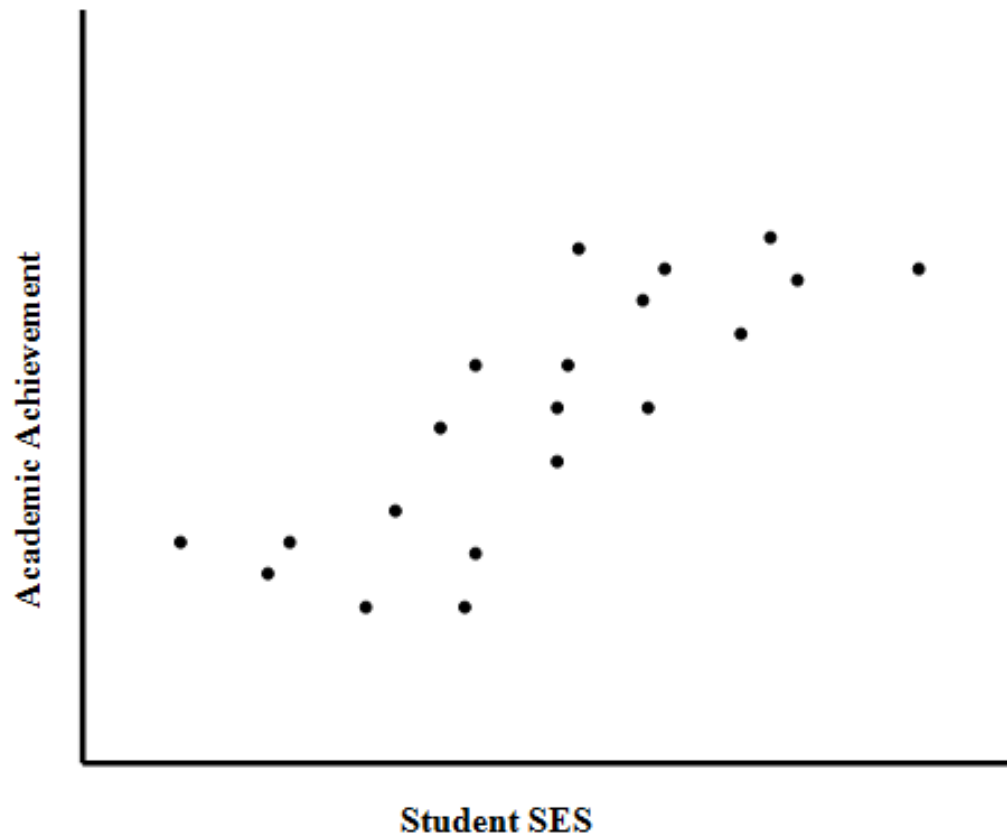
F. Some special applications

- Spatially correlated random effects
- Latent variable models (missing Xs, Xs measured with error)
- Research synthesis – e.g. Meta-analysis

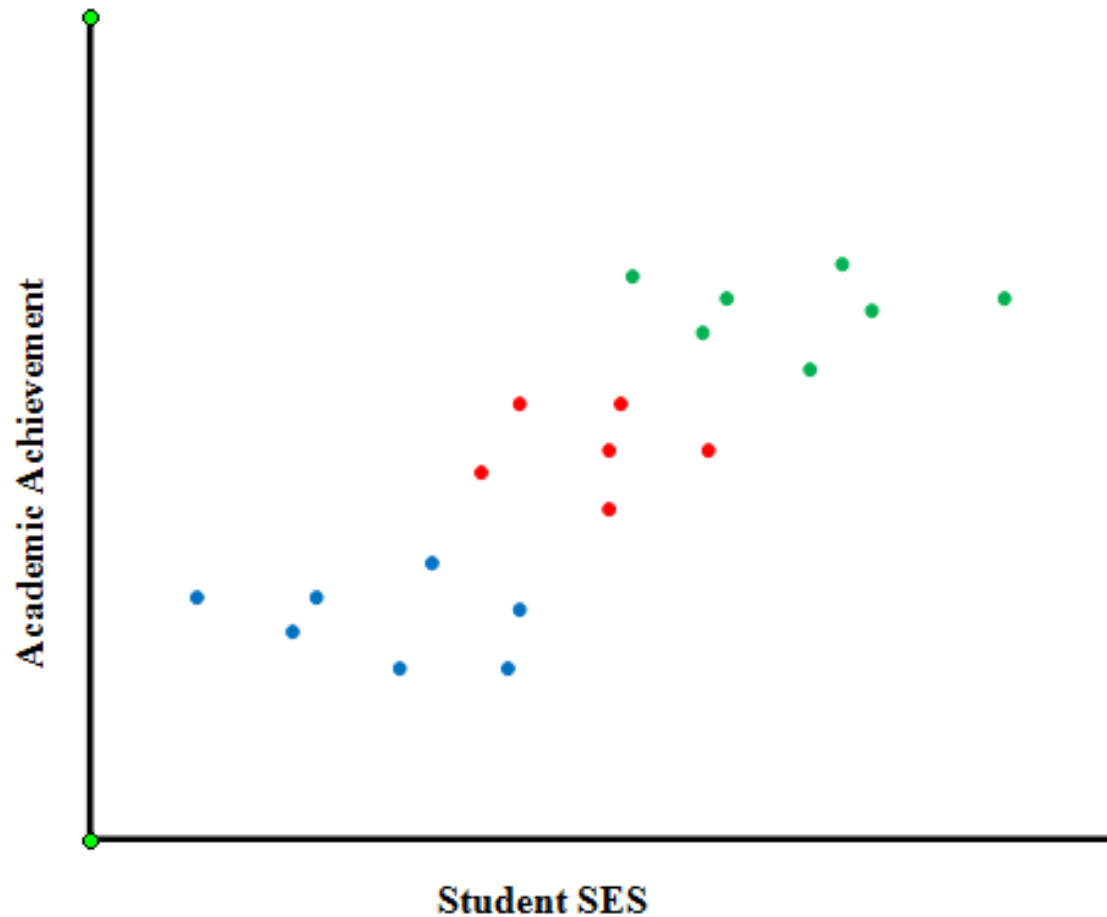
OLS

- ◆ Ordinary least-squares regression (OLS) is a way of analyzing the association between continuous or dichotomous predictor variables and a continuous outcome.
- ◆ Formula $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \varepsilon_i$
 $\varepsilon \text{ iid } \sim N(0, \sigma^2)$
- ◆ Recall that the errors are assumed to be independent of each other – that is, they are uncorrelated.

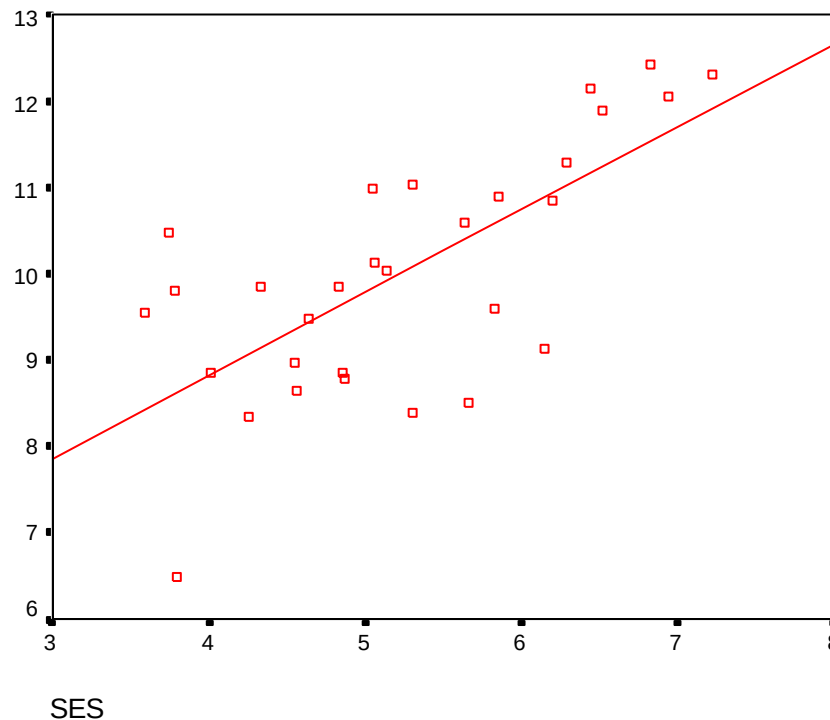
Visual Example – No Clustering



Visual Example – With Clustering

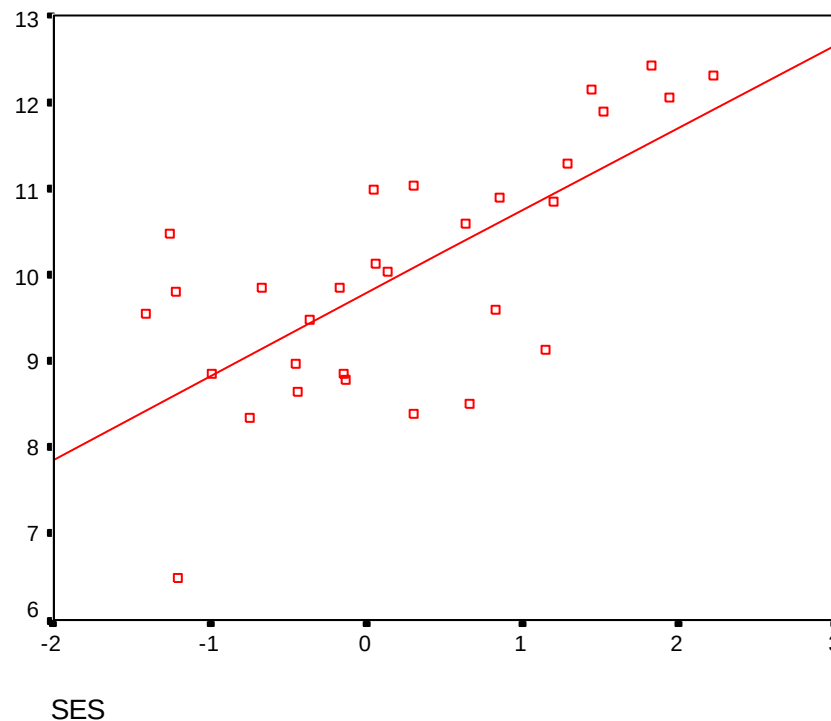


Plot of Math Achievement As a Function of SES in One School



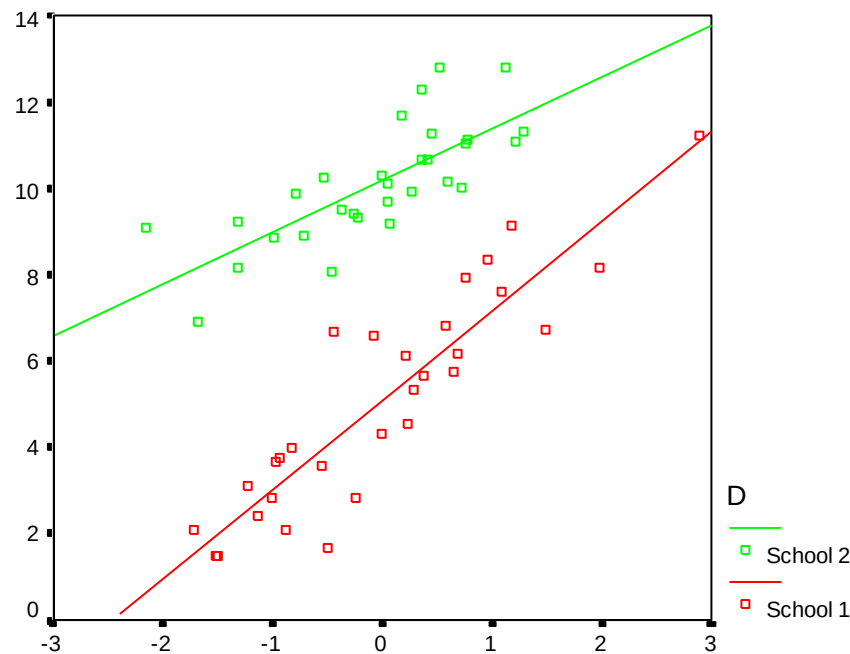
$$\text{MATHACH}_i = \beta_0 + \beta_1(\text{SES}_i) + e_i$$

Plot of MATHACH As a Function of (SES-meanSES) in One School



$$\text{MATHACH}_i = \beta_0 + \beta_1(\text{SES}_i - \text{meanSES}) + e_i$$

Plot of Math Achievement As a Function of SES in Two Schools



$$\text{MATHACH}_{1i}^{\text{SES}} = \beta_{01} + \beta_{11}(\text{SES}_{1i} - \text{meanSES}_1) + e_{1i}$$

$$\text{MATHACH}_{2i} = \beta_{02} + \beta_{12}(\text{SES}_{2i} - \text{meanSES}_2) + e_{2i}$$

Study of SES-Achievement Relationship in J Schools

$$Y_{ij} = \beta_{0j} + \beta_{1j}(X_{ij} - \bar{X}_{\cdot j}) + r_{ij}$$

$$E(\beta_{0j}) = \gamma_0 \quad \text{Var}(\beta_{0j}) = \tau_{00}$$

$$E(\beta_{1j}) = \gamma_1 \quad \text{Var}(\beta_{1j}) = \tau_{11}$$

$$\text{Cov}(\beta_{0j}, \beta_{1j}) = \tau_{01}$$

Developing a Model to Predict β_{0j}, β_{1j}
Within each school

$$Y_{ij} = \beta_{0j} + \beta_{1j}(X_{ij} - \bar{X}_{\cdot j}) + r_{ij}$$

Looking across all schools

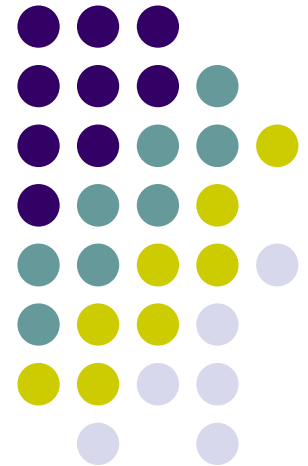
$$\beta_{0j} = \gamma_{00} + \gamma_{01}W_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}W_j + u_{1j}$$

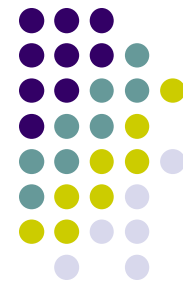
Combined model

$$\begin{aligned} Y_{ij} = & \gamma_{00} + \gamma_{01}W_j \\ & + \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) \\ & + \gamma_{11}W_j(X_{ij} - \bar{X}_{\cdot j}) \\ & + u_{0j} + u_{1j}(X_{ij} - \bar{X}_{\cdot j}) + r_{ij} \end{aligned}$$

Basic HLM Illustrations



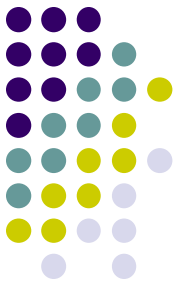
One-Way ANOVA Model



$$\text{Level 1: } Y_{ij} = \beta_{0j} + r_{ij} \quad r_{ij} \sim N(0, \sigma^2)$$

$$\text{Level 2: } \beta_{0j} = \gamma_{00} + u_{0j} \quad u_{0j} \sim N(0, \tau_{00})$$

$$\text{Combined : } Y_{ij} = \gamma_{00} + u_{0j} + r_{ij}$$



Two kinds of intervals common with HLMs

Confidence intervals describe the *precision* of our estimates, how much uncertainty we have about them.

└ 1.96 for a 95% interval
↓

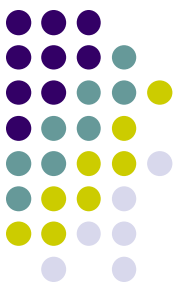
point estimate $\pm (\alpha)$ (standard **error** of estimate)

Plausible value intervals describe the *variation* in our estimates across units. They describe a range values that one might reasonably find for individual units.

└ 1.96 for a 95% interval
↓

point estimate $\pm (\alpha)$ (standard **deviation** of estimate)

Ancillary Statistics: One-Way ANOVA Model



$$95\% \text{ C.I. } (\gamma_{00}) = 12.64 \pm 1.96(.24) = (12.17, 13.11)$$

$$\hat{\rho} = \hat{\tau}_{00} / (\hat{\tau}_{00} + \hat{\sigma}^2) = .18$$

$$\hat{\lambda}_j = \hat{\tau}_{00} / (\hat{\tau}_{00} + \hat{\sigma}^2 / n_j) \quad \hat{\lambda}_{\cdot} = \sum \hat{\lambda}_j / J = .90$$

$$H_0 : \tau_{00} = 0 \Rightarrow \chi^2 \text{ statistic with } J - 1 \text{ df}$$

$$\sum n_j (\bar{Y}_{\cdot j} - \hat{\gamma}_{00})^2 / \hat{\sigma}^2$$

$$\begin{aligned} 95\% P.V.(\beta_{0j}) &= \hat{\gamma}_{00} \pm 1.96(\hat{\tau}_{00})^{1/2} = \\ &= 12.64 \pm 1.96(8.61)^{1/2} = (6.89, 18.39) \end{aligned}$$

“Group – Mean Centering”

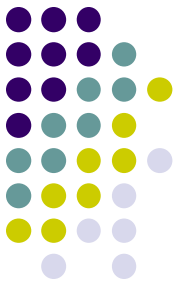
β_{0j} and β_{1j}
are random level – 1
coefficients

Level-2 γ coefficients, also “fixed effects”

$$\gamma_{10} = \text{Ave. Slope}$$

Unconditional variances and covariances

Ancillary Statistics: Random-Coefficient Regression Model



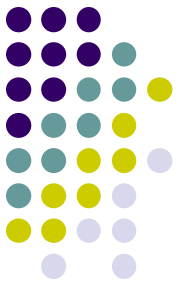
$$\begin{aligned}\text{Proportion of variance explained at level 1} &= \frac{\hat{\sigma}^2(\text{unconditional}) - \hat{\sigma}^2(\text{conditional})}{\hat{\sigma}^2(\text{unconditional})} \\ &= \frac{39.15 - 36.71}{39.15} = 0.062\end{aligned}$$

Reliabilities of:

$$\begin{aligned}\hat{\beta}_{0j} &= .91 \\ \hat{\beta}_{1j} &= .26\end{aligned}$$

$$\begin{aligned}\text{corr}(\hat{\beta}_{0j}, \hat{\beta}_{1j}) &= \hat{\tau}_{01} / (\hat{\tau}_{00} \cdot \hat{\tau}_{11})^{1/2} \\ &= 0.02\end{aligned}$$

Ancillary Statistics: Random-Coefficient Regression Model



- 95% plausible value range for $\beta_{0j} =$

$$12.64 \pm 1.96(2.95) = (6.86, 18.42)$$

- 95% plausible value range for $\beta_{1j} =$

$$2.19 \pm 1.96(0.82) = (0.58, 3.80)$$